

Module 1:

Getting Started Syntax and Data types

Topics:

- Introduction to Python
- How Python Code is Executed
- Numbers
- Strings
- Lists
- Tuples
- Dictionary
- Escape Sequences
- Working with Numbers
- Type Conversion
- Practice and exercise

Module 2:

Control Flow

Topics:

- Ternary Operator
- For Loops and nested Loops
- While Loops
- Break statement
- Continue and Pass Statement
- Iterator and Generator
- Practice and exercise

Module 3:

Functions

Topics:

- Defining Functions
- Arguments
- Types of Functions
- Default Arguments
- xargs and xxargs
- Scope
- Practice and exercise

Module 4:

Data Structures

Topics:

- Lists
- Tuples
- Sets
- Dictionaries
- Lambda Functions and Map Function
- List Comprehensions
- Zip Function
- Swapping Variables
- Strings
- Arrays
- Stacks
- Queues
- Practice and exercise

Module 5:

Errors and Exceptions

Topics:

- Exceptions
- Handling Different Exceptions
- Raising Exceptions
- Practice and exercise

- File methods

- Practice and exercise

Module 8:

Python Modules and Standard Library

Topics:

- Python pip
- `__name__` and `__main__`
- Working With Paths and Directories
- Working with Files
- Working with Zip Files
- Working with CSV Files
- Working with JSON Files
- Working with a SQLite Database
- Working with Timestamps and DateTimes
- Generating Random Values
- Start16- Command-line Arguments
- Start17- Running External Programs

Module 6:

Object Oriented

Topics:

- OOPS Terminology
- Creating Classes
- Instance Objects
- Class Attributes
- Accessing Attributes
- Destroying Objects
- Inheritance
- Overriding
- Overloading

Module 7:

Object Oriented

Topics:

- Opening and Closing Files
- The file Object Attributes
- Reading and Writing Files
- File Positions
- Renaming and Deleting Files

Module 9:

Flask API

Topics:

- Flask Tutorial
- First Flask Application
- Flask App and routing
- Flask URL
- Flask HTTP Methods
- Flask Session

- Flask SQLite and SQLAlchemy
- Flask WTF
- Flask vs Django
- Flask single page app project

Module 10:

Object Oriented

Topics:

- Introduction
- Django Project
- Views
- Models
- Database APIs with sqlite db
- Templates
- Web authentication- a mini project